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Profile Overview

- **Theme** - CYBER
- **Problem Statement Title** - Create A Dark Web Monitoring System
- **Team ID** - 32395993
- **Team Name** - Pratishtha

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IDEA TITLE

➔ Solution Overview

- **Dark Web Scraping** : Automated dark web browsing agent for data collection across multiple pages.
- **DDoS Protection Bypass** : Ensures continuous access to dark web sites by overcoming security measures designed to block automated traffic.
- **AI - Classification** : Stores collected data, dynamically classifies it in real-time through an AI agent, and matches leaked credentials for processing and risk assessment.
- **Automated Email Notification** : Sends alerts for critical issues, ensuring immediate awareness and timely response.
- **Geo Heatmap & Threat Correlation Engine** : Visualizes dark web activity and correlates data to identify threat patterns.
- **Severity Visualizer & Prioritizer** : AI-powered color coding ranks threats by risk to guide swift, effective response.

➔ Problem - Solving

- **Integrated Platform:** Unifies dark web data collection, processing, and analysis in one resilient system.
- **AI-Powered Detection:** Specialized crawlers gather data, fed directly to our AI for intelligent threat identification.

➔ Innovation:

- **Cost-Efficient Architecture:** Designed for low-cost deployment without sacrificing performance.
- **Ultra-Low Latency:** Real-time analysis powered by locally hosted AI — no cloud delay.
- **Intelligent Threat Correlation:** Automatically links related dark web activities, users, and data to reveal hidden connections.

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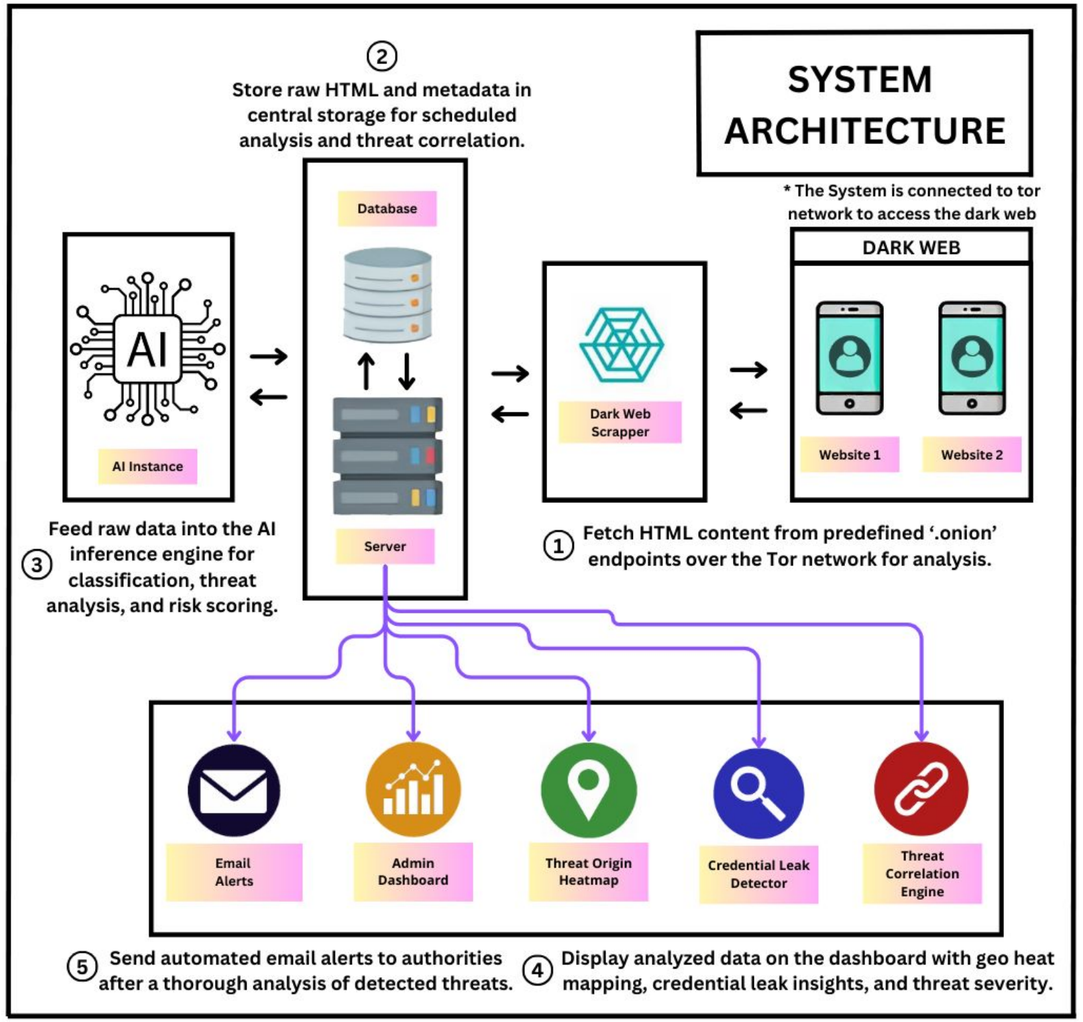
TECHNICAL APPROACH

- ➔ **Backend** : Core technologies are **Flask** and **Docker** and **SQLite** database.
- ➔ **Frontend** : **React JS** and **Tailwind** will be used to make beautiful User Interface for the website.
- ➔ **Hosting** : Local or AWS EC2
- ➔ **AI** : Meta LLama 3.2 - 3b Custom Instruct Model

Tech Stack

Front-End	Back-End	Others
React JS	Python	Tor Network
Tailwind	Flask	Docker
Bootstrap	SQLite3 DB	Meta LLama AI



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FEASIBILITY AND VIABILITY

➔ Feasibility

- Uses web scraping libraries, Tor protocols, and pre-trained NLP models for efficient scraping and Adaptive Content Classification.
- Flask enables rapid development, secure architecture, and easy API integration.
- Deploying in the cloud with GPU support will incur a cost of ₹15,000 per month.
- Locally deployed, it remains free with no API or infrastructure costs.

Cost Comparision

AWS	+	ChatGPT API	=	₹20,000 / mo
AWS	+	Gemini API	=	₹18,000 / mo
Local	+	Local LLM	=	₹0 / mo
GCP	+	Local LLM	=	₹15,000 / mo

➔ Challenges & Risks

- Dark web sites often employ CAPTCHAs, JavaScript challenges, or require logins, complicating automated scraping.
- Dark web marketplaces and forums constantly change names, domains, or branding to evade detection and shutdowns.
- Cyber criminals sometimes hide messages or data dumps within images using steganography, avoiding detection.

➔ Mitigation Strategies

- Automated dark web agent with headless browsing, steganography detection, human - assisted CAPTCHA solving, and adaptive link traversal.

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IMPACT AND BENEFITS

➔ **Target Audience Impact**

- Supports cyber security agencies, government bodies, and law enforcement.
- Delivers real-time, actionable insights for quick decision-making.
- Empowers non-technical users through a user-friendly dashboard.

➔ **Social, Economic & Environmental Impact**

- Reduces cybercrime-related losses for individuals and businesses.
- Helps protect national security and citizen privacy.
- Encourages digital tracking and reporting, reducing paper-based operations.

➔ **Long-Term Value**

- Easily scalable across regions and languages.
- Modular architecture allows future feature integration.
- AI model can evolve to detect emerging threat types.
- Maps cybercrime origins for better global response.
- Can be expanded to mobile, browser extensions, or APIs.

➔ **Proactive Threat Intelligence**

- Real-time monitoring helps prevent attacks before damage occurs.
- Risk scoring prioritizes critical threats for faster action.
- Enables pattern recognition for long-term trend analysis.

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